

Worked examples for Lecture 12: Finite elasticity

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Michaelmas Term, 2026

These examples accompany Lecture 12. They are intended as practice in the concrete calculations that underlie the lecture, and are less demanding than the questions on the problem sets. Numerical results were checked with the scripts in the python directory.

Example 1: Simple shear. A body with referential density $\rho(\mathbf{x})$ undergoes the motion

$$\varphi(\mathbf{x}, t) = (x_1 + \gamma t x_2, x_2, x_3),$$

where γ is a constant shear rate, and the reference body is the position of the body at $t = 0$. (a) Compute the referential velocity, the deformation gradient and the Jacobian, and show that the motion preserves volume. (b) Find the inverse motion $\varphi^{-1}(\cdot, t)$, and hence the spatial density and the spatial velocity field, i.e. the velocity of whichever particle occupies the spatial point $\mathbf{y}$ at time t . Compare these with their referential counterparts. (c) Compute $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, show that $\delta \mathbf{x} \cdot \mathbf{C} \delta \mathbf{x}$ is the squared length of the image of a small material line element $\delta \mathbf{x}$, and find the eigenvalues and eigenvectors of $\mathbf{C}$. The square roots of the eigenvalues are the *principal stretches*.

Solution. (a) Differentiating the motion with respect to time at fixed $\mathbf{x}$, and with respect to $\mathbf{x}$ at fixed t , we obtain

$$\mathbf{v}(\mathbf{x}, t) = (\gamma x_2, 0, 0), \quad \mathbf{F}(\mathbf{x}, t) = \begin{pmatrix} 1 & \gamma t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (1)$$

The deformation gradient is upper triangular with unit diagonal, and so

$$J = \det \mathbf{F} = 1 \quad (2)$$

for all $\mathbf{x}$ and t . Applying the change of variables formula with $f = 1$ to any sub-body U , we find that the volume of U_t equals that of U , so the motion is *isochoric*. Mass conservation, $\rho(\mathbf{x}) = J \varrho[\varphi(\mathbf{x}, t), t]$, then gives $\varrho[\varphi(\mathbf{x}, t), t] = \rho(\mathbf{x})$: each particle retains its initial density.

(b) Setting $\mathbf{y} = \varphi(\mathbf{x}, t)$ and solving for $\mathbf{x}$ gives the inverse motion

$$\varphi^{-1}(\mathbf{y}, t) = (y_1 - \gamma t y_2, y_2, y_3), \quad (3)$$

which is well-defined for all t , as it must be since $J \neq 0$. The spatial density is therefore

$$\varrho(\mathbf{y}, t) = \rho(y_1 - \gamma t y_2, y_2, y_3). \quad (4)$$

Although $J = 1$, the spatial density at a fixed point $\mathbf{y}$ is in general time-dependent. For example, if $\rho(\mathbf{x}) = \rho_0(1 + \alpha x_1)$, then $\rho(\mathbf{y}, t) = \rho_0[1 + \alpha(y_1 - \gamma t y_2)]$, because particles of differing density are carried past the point $\mathbf{y}$ by the shear. Only for uniform ρ is ρ constant in time. For the velocity, substituting eq. (3) into eq. (1) gives the spatial velocity field

$$\mathbf{v}[\varphi^{-1}(\mathbf{y}, t), t] = (\gamma y_2, 0, 0). \quad (5)$$

For this particular motion the spatial and referential velocity fields have the same functional form, because the velocity depends only on x_2 , and $x_2 = y_2$ is unchanged by the motion. Their meanings are nevertheless different: eq. (1) gives the velocity of the particle labelled $\mathbf{x}$, while eq. (5) gives the velocity observed at the fixed spatial point $\mathbf{y}$. Example 4 provides a case in which the two fields differ in form.

(c) From eq. (1), and writing $a = \gamma t$ for brevity,

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} = \begin{pmatrix} 1 & a & 0 \\ a & 1 + a^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (6)$$

By the Taylor expansion of the motion, a small material line element $\delta \mathbf{x}$ at $\mathbf{x}$ is mapped to $\mathbf{F} \delta \mathbf{x}$, whose squared length is

$$\|\mathbf{F} \delta \mathbf{x}\|^2 = \delta \mathbf{x} \cdot \mathbf{F}^T \mathbf{F} \delta \mathbf{x} = \delta \mathbf{x} \cdot \mathbf{C} \delta \mathbf{x}. \quad (7)$$

Hence $\mathbf{C}$ is symmetric and positive-definite, and a unit line element along an eigenvector of $\mathbf{C}$ with eigenvalue λ has its length multiplied by $\sqrt{\lambda}$. The x_3 -direction is an eigenvector with eigenvalue 1. In the (x_1, x_2) -plane the characteristic equation is

$$\lambda^2 - (2 + a^2)\lambda + 1 = 0, \quad (8)$$

with roots

$$\lambda_{\pm} = 1 + \frac{1}{2}a^2 \pm a\sqrt{1 + \frac{1}{4}a^2}. \quad (9)$$

The product of the roots is $\lambda_+ \lambda_- = 1 = \det \mathbf{C} = J^2$, as it must be for an isochoric motion. The principal stretches take the simple form

$$\sqrt{\lambda_{\pm}} = \sqrt{1 + \frac{1}{4}a^2} \pm \frac{1}{2}a, \quad (10)$$

as may be verified by squaring the right-hand side. The eigenvector equation $(1 - \lambda)u_1 + au_2 = 0$ gives $\mathbf{u} \propto (1, (\lambda - 1)/a, 0)$, and using eqs. (9) and (10) we have $(\lambda_{\pm} - 1)/a = \frac{1}{2}a \pm \sqrt{1 + \frac{1}{4}a^2} = \pm \sqrt{\lambda_{\pm}}$, so that

$$\mathbf{u}_{\pm} \propto (1, \pm \sqrt{\lambda_{\pm}}, 0). \quad (11)$$

These are orthogonal, since $\mathbf{u}_+ \cdot \mathbf{u}_- = 1 - \sqrt{\lambda_+ \lambda_-} = 0$. For small a we have $\lambda_{\pm} = 1 \pm a + O(a^2)$, principal stretches $1 \pm \frac{1}{2}a$, and eigenvectors tending to $(1, \pm 1, 0)/\sqrt{2}$: to first order, simple shear is an extension along one diagonal of the (x_1, x_2) -plane and an equal contraction along the other, which is the familiar statement of small-strain theory. For large a the largest stretch is approximately a , and its direction $(1, \sqrt{\lambda_+}, 0)$ tends towards the x_2 -axis, since material lines initially along x_2 are drawn out most strongly by the shear.

Example 2: Polar decomposition by hand. Consider the plane deformation gradient

$$\mathbf{F} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix},$$

the third direction being trivial ($F_{33} = 1, F_{13} = F_{23} = F_{31} = F_{32} = 0$). The polar decomposition theorem of Lecture 13 states that $\mathbf{F} = \mathbf{R}\mathbf{U} = \mathbf{V}\mathbf{R}$ with $\mathbf{R}$ a rotation matrix and $\mathbf{U}, \mathbf{V}$ symmetric and positive-definite. Compute $\mathbf{C} = \mathbf{F}^T\mathbf{F}$ and its eigenvalues and eigenvectors; construct $\mathbf{U} = \sqrt{\mathbf{C}}$ from the eigen-decomposition; show that $\mathbf{R} = \mathbf{F}\mathbf{U}^{-1}$ is a rotation and find its angle; compute $\mathbf{V} = \mathbf{R}\mathbf{U}^T$; and describe the image of the unit circle.

Solution. Direct multiplication gives

$$\mathbf{C} = \mathbf{F}^T\mathbf{F} = \begin{pmatrix} 2 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}, \quad (12)$$

with $\text{tr } \mathbf{C} = 6$ and $\det \mathbf{C} = 4 = (\det \mathbf{F})^2$. The characteristic equation $\lambda^2 - 6\lambda + 4 = 0$ has roots

$$\lambda_{\pm} = 3 \pm \sqrt{5}, \quad (13)$$

and the eigenvector equation $(4 - \lambda)u_1 + 2u_2 = 0$ gives $\mathbf{u} \propto (2, \lambda - 4)$, so that

$$\mathbf{u}_{\pm} \propto (2, -1 \pm \sqrt{5}), \quad \|\mathbf{u}_{\pm}\|^2 = 10 \mp 2\sqrt{5}. \quad (14)$$

These are orthogonal, as $4 - (\sqrt{5} - 1)(\sqrt{5} + 1) = 0$. The eigenvector $\mathbf{u}_+$ makes an angle $\alpha = \arctan[(\sqrt{5} - 1)/2] \approx 31.7^\circ$ with the x_1 -axis. The square roots of the eigenvalues have the closed forms

$$\sqrt{\lambda_{\pm}} = \frac{\sqrt{5} \pm 1}{\sqrt{2}}, \quad (15)$$

since $(\sqrt{5} \pm 1)^2/2 = (6 \pm 2\sqrt{5})/2 = 3 \pm \sqrt{5}$; these are the principal stretches, with product $2 = \det \mathbf{F}$.

Writing $\mathbf{q}_{\pm} = \mathbf{u}_{\pm}/\|\mathbf{u}_{\pm}\|$, the square root of $\mathbf{C}$ is

$$\mathbf{U} = \sqrt{\lambda_+} \mathbf{q}_+ \mathbf{q}_+^T + \sqrt{\lambda_-} \mathbf{q}_- \mathbf{q}_-^T. \quad (16)$$

Using $1/(10 - 2\sqrt{5}) = (5 + \sqrt{5})/40$, the first projector is

$$\mathbf{q}_+ \mathbf{q}_+^T = \frac{5 + \sqrt{5}}{40} \begin{pmatrix} 4 & 2\sqrt{5} - 2 \\ 2\sqrt{5} - 2 & 6 - 2\sqrt{5} \end{pmatrix} = \begin{pmatrix} \frac{5+\sqrt{5}}{10} & \frac{\sqrt{5}}{5} \\ \frac{\sqrt{5}}{5} & \frac{5-\sqrt{5}}{10} \end{pmatrix}, \quad (17)$$

and $\mathbf{q}_- \mathbf{q}_-^T = \mathbf{1} - \mathbf{q}_+ \mathbf{q}_+^T$. Substituting into eq. (16) with eq. (15), the entries of $\mathbf{U}$ are

$$\begin{aligned} U_{11} &= \frac{(\sqrt{5} + 1)(5 + \sqrt{5}) + (\sqrt{5} - 1)(5 - \sqrt{5})}{10\sqrt{2}} = \frac{12\sqrt{5}}{10\sqrt{2}} = \frac{6}{\sqrt{10}}, \\ U_{22} &= \frac{(\sqrt{5} + 1)(5 - \sqrt{5}) + (\sqrt{5} - 1)(5 + \sqrt{5})}{10\sqrt{2}} = \frac{8\sqrt{5}}{10\sqrt{2}} = \frac{4}{\sqrt{10}}, \\ U_{12} &= \frac{\sqrt{5}}{5} (\sqrt{\lambda_+} - \sqrt{\lambda_-}) = \frac{2}{\sqrt{10}}. \end{aligned} \quad (18)$$

Hence

$$\mathbf{U} = \frac{1}{\sqrt{10}} \begin{pmatrix} 6 & 2 \\ 2 & 4 \end{pmatrix} = \sqrt{\frac{2}{5}} \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \quad (19)$$

and we check that $\mathbf{U}^2 = \frac{2}{5} \begin{pmatrix} 10 & 5 \\ 5 & 5 \end{pmatrix} = \mathbf{C}$, $\text{tr } \mathbf{U} = \sqrt{10} = \sqrt{\lambda_+} + \sqrt{\lambda_-}$ and $\det \mathbf{U} = 2 = \det \mathbf{F}$. (In two dimensions there is a shortcut: the Cayley–Hamilton theorem gives $\mathbf{U}^2 - (\text{tr } \mathbf{U})\mathbf{U} + (\det \mathbf{U})\mathbf{1} = \mathbf{0}$, so $\mathbf{U} = (\mathbf{C} + \sqrt{\det \mathbf{C}}\mathbf{1})/\text{tr } \mathbf{U}$ with $(\text{tr } \mathbf{U})^2 = \text{tr } \mathbf{C} + 2\sqrt{\det \mathbf{C}} = 10$, which reproduces eq. (19) immediately.)

Inverting the 2×2 matrix,

$$\mathbf{U}^{-1} = \frac{1}{\det \mathbf{U}} \begin{pmatrix} U_{22} & -U_{12} \\ -U_{12} & U_{11} \end{pmatrix} = \frac{1}{\sqrt{10}} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix}, \quad (20)$$

and therefore

$$\mathbf{R} = \mathbf{F}\mathbf{U}^{-1} = \frac{1}{\sqrt{10}} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ -1 & 3 \end{pmatrix} = \frac{1}{\sqrt{10}} \begin{pmatrix} 3 & 1 \\ -1 & 3 \end{pmatrix}. \quad (21)$$

The columns of $\mathbf{R}$ are orthonormal, so $\mathbf{R}^T \mathbf{R} = \mathbf{1}$, and $\det \mathbf{R} = (9 + 1)/10 = 1$; hence $\mathbf{R}$ is a rotation. Comparing with $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ we read off $\cos \theta = 3/\sqrt{10}$ and $\sin \theta = -1/\sqrt{10}$, so that

$$\tan \theta = -\frac{1}{3}, \quad \theta \approx -18.4^\circ, \quad (22)$$

a clockwise rotation. The left stretch is most easily found from $\mathbf{V} = \mathbf{R}\mathbf{U}\mathbf{R}^T = \mathbf{F}\mathbf{R}^T$:

$$\mathbf{V} = \frac{1}{\sqrt{10}} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix} = \frac{1}{\sqrt{10}} \begin{pmatrix} 7 & 1 \\ 1 & 3 \end{pmatrix}, \quad (23)$$

which is symmetric with $\text{tr } \mathbf{V} = \sqrt{10}$ and $\det \mathbf{V} = 2$, so it has the same eigenvalues as $\mathbf{U}$, and $\mathbf{V}^2 = \begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix} = \mathbf{F}\mathbf{F}^T$. As a final check,

$$\mathbf{V}\mathbf{R} = \frac{1}{10} \begin{pmatrix} 7 & 1 \\ 1 & 3 \end{pmatrix} \begin{pmatrix} 3 & 1 \\ -1 & 3 \end{pmatrix} = \frac{1}{10} \begin{pmatrix} 20 & 10 \\ 0 & 10 \end{pmatrix} = \mathbf{F}, \quad (24)$$

and similarly $\mathbf{R}\mathbf{U} = \mathbf{F}$. The eigenvectors of $\mathbf{V}$ are those of $\mathbf{F}\mathbf{F}^T$, namely $\mathbf{w}_\pm \propto (1, -2 \pm \sqrt{5})$, and one verifies that $\mathbf{R}\mathbf{u}_+ \propto \mathbf{w}_+$: the eigenvectors of $\mathbf{V}$ are the rotated eigenvectors of $\mathbf{U}$. Indeed $\mathbf{w}_+$ makes an angle $\beta = \arctan(\sqrt{5} - 2) \approx 13.3^\circ = \alpha + \theta$ with the x_1 -axis.

Geometrically (Fig. 1), $\mathbf{F}$ maps the unit circle to an ellipse with semi-axes $\sqrt{\lambda_\pm} \approx 2.29$ and 0.87 along $\mathbf{w}_\pm$, whose area $\pi\sqrt{\lambda_+\lambda_-} = 2\pi$ is J times that of the circle. The decomposition $\mathbf{F} = \mathbf{R}\mathbf{U}$ realises this as a stretch along the fixed directions $\mathbf{q}_\pm$ followed by the rotation, while $\mathbf{F} = \mathbf{V}\mathbf{R}$ rotates first and then stretches along the co-rotated directions $\mathbf{w}_\pm = \mathbf{R}\mathbf{q}_\pm$. Note that this matrix is the deformation gradient of the simple shear of Example 1 with $\gamma t = 1$, composed with a stretch by a factor of 2 along x_1 ; neither ingredient is a rotation, yet the deformation contains a rotation of 18.4° that only the polar decomposition makes visible.

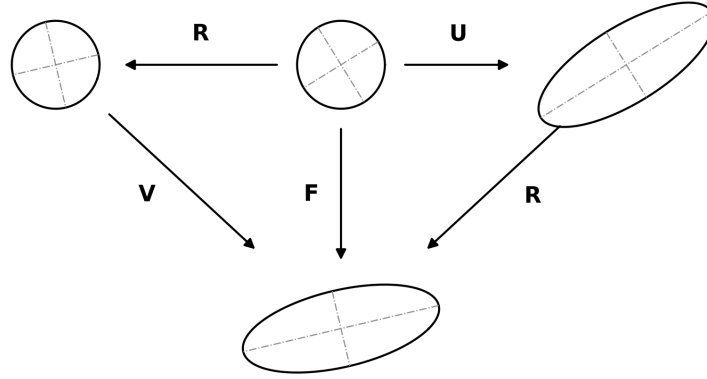


Fig. 1: The polar decomposition of the deformation gradient of Example 2, drawn in the manner of the corresponding figure in Lecture 13. The unit circle (centre of the upper row) is stretched by $\mathbf{U}$ along its eigenvectors (dash-dotted lines, at 31.7° and 121.7°) into an ellipse with semi-axes 2.29 and 0.87, and rotated by $\mathbf{R}$ through -18.4° . Either order of operations produces the same final ellipse, whose axes lie along the eigenvectors of $\mathbf{V}$.

Example 3: Rigid body motions. Let $\varphi(\mathbf{x}, t) = \mathbf{a}(t) + \mathbf{Q}(t)\mathbf{x}$, with $\mathbf{a}(t)$ a translation and $\mathbf{Q}(t)$ a rotation matrix. (a) Show that $\mathbf{F} = \mathbf{Q}$, $J = 1$ and $\mathbf{C} = \mathbf{1}$. (b) Show that, if the origin of the reference body is placed at its centre of mass, the kinetic energy is $\frac{1}{2}M\|\dot{\mathbf{a}}\|^2$ plus a rotational term, and identify the latter. (c) Show that any strain energy function satisfying material frame indifference has $W(\mathbf{x}, \mathbf{Q}) = W(\mathbf{x}, \mathbf{1})$, so that the potential energy is unchanged by the motion. (d) Determine which of $W_1 = \frac{1}{2}\mu \operatorname{tr}(\mathbf{F}^T \mathbf{F} - \mathbf{1})$, $W_2 = \frac{1}{2}\mu \operatorname{tr}(\mathbf{F} - \mathbf{1})$ and $W_3 = \frac{1}{2}\mu \|\mathbf{F} - \mathbf{1}\|^2$, where $\|\mathbf{A}\|^2 = A_{ij}A_{ij}$, are frame indifferent, giving an explicit rotation $\mathbf{Q}$ for those that are not.

Solution. (a) Since $\varphi_i = a_i(t) + Q_{ij}(t)x_j$, we have $F_{ij} = \partial\varphi_i/\partial x_j = Q_{ij}$. A rotation matrix satisfies $\mathbf{Q}^T \mathbf{Q} = \mathbf{1}$ and $\det \mathbf{Q} = 1$, so $J = 1$ and $\mathbf{C} = \mathbf{Q}^T \mathbf{Q} = \mathbf{1}$. By eq. (7), no material line element changes its length, which is what we mean by a rigid motion.

(b) The referential velocity is $\mathbf{v} = \dot{\mathbf{a}} + \dot{\mathbf{Q}}\mathbf{x}$. Differentiating $\mathbf{Q}^T \mathbf{Q} = \mathbf{1}$ gives $\dot{\mathbf{Q}}^T \mathbf{Q} + \mathbf{Q}^T \dot{\mathbf{Q}} = \mathbf{0}$, so the matrix $\boldsymbol{\Omega} = \mathbf{Q}^T \dot{\mathbf{Q}}$ is antisymmetric and can be written $\Omega_{ij} = -\epsilon_{ijk}\omega_k$ for some vector $\boldsymbol{\omega}(t)$, the angular velocity referred to the reference body. Then $\boldsymbol{\Omega}\mathbf{x} = \boldsymbol{\omega} \times \mathbf{x}$ and $\dot{\mathbf{Q}}\mathbf{x} = \mathbf{Q}(\boldsymbol{\omega} \times \mathbf{x})$, so that

$$\|\mathbf{v}\|^2 = \|\dot{\mathbf{a}}\|^2 + 2\dot{\mathbf{a}} \cdot \mathbf{Q}(\boldsymbol{\omega} \times \mathbf{x}) + \|\boldsymbol{\omega} \times \mathbf{x}\|^2, \quad (25)$$

where we used that $\mathbf{Q}$ preserves lengths. Integrating against ρ over M , the middle term contains $\int_M \rho \mathbf{x} d^3\mathbf{x}$, which vanishes when the origin is at the centre of mass. Using $\|\boldsymbol{\omega} \times \mathbf{x}\|^2 = \|\boldsymbol{\omega}\|^2 \|\mathbf{x}\|^2 - (\boldsymbol{\omega} \cdot \mathbf{x})^2$, the kinetic energy becomes

$$\mathcal{T} = \frac{1}{2}M\|\dot{\mathbf{a}}\|^2 + \frac{1}{2}\omega_i I_{ij} \omega_j, \quad I_{ij} = \int_M \rho(\mathbf{x}) (\|\mathbf{x}\|^2 \delta_{ij} - x_i x_j) d^3\mathbf{x}, \quad (26)$$

where $M = \int_M \rho d^3\mathbf{x}$ is the total mass and I_{ij} is the inertia tensor of the reference body, which is constant in time. This is the familiar decomposition of the kinetic energy of

a rigid body into translational and rotational parts, here obtained directly from the referential formula of the lecture.

(c) Material frame indifference requires $W(\mathbf{x}, \mathbf{QF}) = W(\mathbf{x}, \mathbf{F})$ for all deformation gradients and all rotations. Setting $\mathbf{F} = \mathbf{1}$ gives $W(\mathbf{x}, \mathbf{Q}) = W(\mathbf{x}, \mathbf{1})$, and hence for the rigid motion

$$\mathcal{V}(t) = \int_M W[\mathbf{x}, \mathbf{Q}(t)] d^3\mathbf{x} = \int_M W(\mathbf{x}, \mathbf{1}) d^3\mathbf{x}, \quad (27)$$

which is independent of time. The Lagrangian therefore reduces, up to a constant, to the kinetic energy of eq. (26), and Hamilton's principle would return the equations of motion of a free rigid body.

(d) For W_1 we have $W_1(\mathbf{QF}) = \frac{1}{2}\mu \operatorname{tr}(\mathbf{F}^T \mathbf{Q}^T \mathbf{QF} - \mathbf{1}) = W_1(\mathbf{F})$ for every rotation, so W_1 is frame indifferent; it depends on $\mathbf{F}$ only through $\mathbf{C}$. For the other two it suffices to exhibit a single counterexample. Take the rotation by $\pi/2$ about the x_3 -axis,

$$\mathbf{Q} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \operatorname{tr} \mathbf{Q} = 1, \quad (28)$$

and compare $\mathbf{F} = \mathbf{1}$ with $\mathbf{F} = \mathbf{Q}$. We find $W_2(\mathbf{1}) = 0$ but $W_2(\mathbf{Q}) = \frac{1}{2}\mu(1 - 3) = -\mu$, while $W_3(\mathbf{1}) = 0$ but

$$W_3(\mathbf{Q}) = \frac{1}{2}\mu(Q_{ij} - \delta_{ij})(Q_{ij} - \delta_{ij}) = \frac{1}{2}\mu(Q_{ij}Q_{ij} - 2Q_{ii} + 3) = \mu(3 - \operatorname{tr} \mathbf{Q}) = 2\mu. \quad (29)$$

Neither W_2 nor W_3 is frame indifferent: both assign a change of potential energy to a rigid rotation of the undeformed body. More generally, for a rotation through an angle ψ about any axis $\operatorname{tr} \mathbf{Q} = 1 + 2\cos \psi$, so $W_3(\mathbf{Q}) = 2\mu(1 - \cos \psi)$. The function W_3 is what one obtains by naively treating the full displacement gradient, rather than its symmetric part, as a measure of strain, and this calculation shows why the linearised theory of the next lecture must be built from the symmetric strain tensor. Note finally that frame indifference is necessary but not sufficient for a physically sensible strain energy: W_1 passes the test, yet $\partial W_1 / \partial F_{ij} = \mu F_{ij}$ does not vanish at $\mathbf{F} = \mathbf{1}$, so a body with this strain energy is not stress-free in its reference configuration.

Example 4: Uniform stretch of a cube. Let the reference body M be the unit cube $[0, 1]^3$ with uniform referential density ρ , and consider the motion $\varphi(\mathbf{x}, t) = (\lambda_1 x_1, \lambda_2 x_2, \lambda_3 x_3)$ with $\lambda_i > 0$. (a) Compute the mass, the volume of M_t and the spatial density. (b) Now let $\lambda_i = \lambda_i(t)$. Compute the kinetic energy from the referential formula of the lecture, and again as an integral over M_t of $\frac{1}{2}\varrho \|\mathbf{v}\|^2$ with the spatial velocity field, and check that the two agree.

Solution. (a) The deformation gradient is $\mathbf{F} = \operatorname{diag}(\lambda_1, \lambda_2, \lambda_3)$ with $J = \lambda_1 \lambda_2 \lambda_3 > 0$. The instantaneous body is the box $M_t = [0, \lambda_1] \times [0, \lambda_2] \times [0, \lambda_3]$, whose volume is $\lambda_1 \lambda_2 \lambda_3 = J$, in agreement with the change of variables formula $\int_{M_t} d^3\mathbf{y} = \int_M J d^3\mathbf{x}$. The mass is $\int_M \rho d^3\mathbf{x} = \rho$, and mass conservation gives the spatial density

$$\varrho(\mathbf{y}, t) = \frac{\rho}{J} = \frac{\rho}{\lambda_1 \lambda_2 \lambda_3}, \quad (30)$$

which is uniform, and satisfies $\int_{M_t} \varrho d^3\mathbf{y} = (\rho/J)J = \rho$ as it must.

(b) The referential velocity is $v_i = \dot{\lambda}_i x_i$ (no sum on i), so the kinetic energy is

$$\mathcal{T} = \frac{1}{2} \rho \int_M \sum_{i=1}^3 \dot{\lambda}_i^2 x_i^2 d^3 \mathbf{x} = \frac{1}{2} \rho \sum_{i=1}^3 \dot{\lambda}_i^2 \int_0^1 x_i^2 dx_i = \frac{\rho}{6} (\dot{\lambda}_1^2 + \dot{\lambda}_2^2 + \dot{\lambda}_3^2). \quad (31)$$

For the spatial calculation, the particle at $\mathbf{y} \in M_t$ is $x_i = y_i / \lambda_i$, so the spatial velocity field is

$$v_i[\varphi^{-1}(\mathbf{y}, t), t] = \frac{\dot{\lambda}_i}{\lambda_i} y_i, \quad (32)$$

which, in contrast to Example 1, differs in form from the referential field. Using eq. (30) and $\int_{M_t} y_1^2 d^3 \mathbf{y} = \frac{1}{3} \lambda_1^3 \lambda_2 \lambda_3 = \frac{1}{3} J \lambda_1^2$ (and similarly for y_2, y_3),

$$\frac{1}{2} \int_{M_t} \rho \|\mathbf{v}\|^2 d^3 \mathbf{y} = \frac{\rho}{2J} \sum_{i=1}^3 \frac{\dot{\lambda}_i^2}{\lambda_i^2} \cdot \frac{J \lambda_i^2}{3} = \frac{\rho}{6} \sum_{i=1}^3 \dot{\lambda}_i^2, \quad (33)$$

in agreement with eq. (31). The two calculations are related by the change of variables $\mathbf{y} = \varphi(\mathbf{x}, t)$, under which $\rho d^3 \mathbf{y} = \rho d^3 \mathbf{x}$; the referential form is the more convenient because its domain of integration does not move.

Example 5: Hamilton's principle for an elastic rod. A thin rod of unit cross-sectional area occupies $x \in [0, L]$ in its reference configuration and has uniform referential density ρ . Its motion is a scalar function $\varphi(x, t)$, with $v = \partial \varphi / \partial t$ and $F = \partial \varphi / \partial x$, and its strain energy function is $W(F) = \frac{1}{2} E (F - 1)^2$ with $E > 0$ a constant. Starting from the action with Lagrangian density $\frac{1}{2} \rho v^2 - W(F)$, derive the equation of motion and the natural boundary conditions at $x = 0$ and $x = L$. Identify the first Piola–Kirchhoff stress. Then linearise about the reference configuration by writing $\varphi = x + su$ and obtain the wave equation and the wave speed.

Solution. The action is

$$\mathcal{S}(\varphi) = \int_0^T \int_0^L \left\{ \frac{1}{2} \rho \left(\frac{\partial \varphi}{\partial t} \right)^2 - \frac{1}{2} E \left(\frac{\partial \varphi}{\partial x} - 1 \right)^2 \right\} dx dt. \quad (34)$$

For a variation $\delta \varphi$ vanishing at $t = 0$ and $t = T$, the first variation is

$$\delta \mathcal{S} = \int_0^T \int_0^L \left\{ \rho \frac{\partial \varphi}{\partial t} \frac{\partial \delta \varphi}{\partial t} - T \frac{\partial \delta \varphi}{\partial x} \right\} dx dt, \quad T = \frac{dW}{dF} = E \left(\frac{\partial \varphi}{\partial x} - 1 \right). \quad (35)$$

Integrating the first term by parts in time, the boundary contribution vanishes by the fixed end-point condition, while the second term is integrated by parts in x :

$$\delta \mathcal{S} = \int_0^T \int_0^L \left\{ -\rho \frac{\partial^2 \varphi}{\partial t^2} + \frac{\partial T}{\partial x} \right\} \delta \varphi dx dt - \int_0^T [T \delta \varphi]_{x=0}^{x=L} dt. \quad (36)$$

For $\delta \mathcal{S}$ to vanish for all admissible variations, the volume and boundary terms must vanish separately. Choosing $\delta \varphi$ supported in the interior gives the Euler–Lagrange equation

$$\rho \frac{\partial^2 \varphi}{\partial t^2} = \frac{\partial T}{\partial x} = E \frac{\partial^2 \varphi}{\partial x^2}, \quad (37)$$

and then choosing $\delta\varphi$ with arbitrary values at the ends gives the natural boundary conditions $T(0, t) = T(L, t) = 0$, i.e.

$$\frac{\partial\varphi}{\partial x} = 1 \quad \text{at} \quad x = 0 \quad \text{and} \quad x = L. \quad (38)$$

Comparing with the lecture, $T = \partial W / \partial F$ is the one-dimensional first Piola–Kirchhoff stress, eq. (37) is $\rho \partial v / \partial t = \partial T / \partial x$, and eq. (38) is the traction-free condition $T\hat{n} = 0$ with $\hat{n} = \pm 1$ the outward normal at the two ends. Physically, an element with $F > 1$ is stretched and carries the tension $E(F - 1)$, an element with $F < 1$ is compressed, and the free ends of the rod carry no force. Had one end been held fixed instead, say $\varphi(0, t) = 0$, the admissible variations would satisfy $\delta\varphi(0, t) = 0$, no boundary term would arise there, and the condition at that end would be the prescribed (essential) one rather than a natural one.

Linearising, set $\varphi(x, t) = x + su(x, t)$ with s small. Then $F = 1 + s \partial u / \partial x$, $T = sE \partial u / \partial x$, and eq. (37) becomes, after dividing by s ,

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad c = \sqrt{\frac{E}{\rho}}, \quad (39)$$

with $\partial u / \partial x = 0$ at the free ends. The normal modes are $u = \cos(n\pi x / L) \cos(\omega_n t)$ with $\omega_n = n\pi c / L$. Because W is quadratic in F the exact equation (37) was already linear, so here the linearisation is exact. This cannot happen in three dimensions: a strain energy that is quadratic in $\mathbf{F}$ and frame indifferent must be a linear function of $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, and then $\partial W / \partial F_{ij}$ is linear in $\mathbf{F}$ and non-zero at $\mathbf{F} = \mathbf{1}$ (as for W_1 in Example 3), so the reference configuration is not stress-free. The linearisation carried out in Lecture 13 is therefore a genuine approximation.

Example 6: Traction as force per unit reference area. Consider again the uniform stretch of Example 4, and suppose that the strain energy function is $W(\mathbf{F}) = \frac{1}{2}\kappa(J - 1)^2$ with $\kappa > 0$ constant and $J = \det \mathbf{F}$. (a) Show that $\partial J / \partial F_{ij} = J F_{ji}^{-1}$, and hence compute the first Piola–Kirchhoff stress. (b) Compute the traction $\mathbf{t} = \mathbf{T}\hat{\mathbf{n}}$ on the face $x_1 = 1$ of the reference cube, the total force acting on the image of this face, and the corresponding force per unit *current* area (the Cauchy traction). Show that the two tractions differ by the factor $\lambda_2 \lambda_3$.

Solution. (a) We first note that W is frame indifferent, since $\det(\mathbf{QF}) = \det \mathbf{Q} \det \mathbf{F} = J$. To differentiate the determinant, let $\delta\mathbf{F}$ be a small perturbation and write $\mathbf{F} + \delta\mathbf{F} = \mathbf{F}(\mathbf{1} + \mathbf{F}^{-1}\delta\mathbf{F})$. For any matrix $\mathbf{A}$, expanding the determinant shows that $\det(\mathbf{1} + \mathbf{A}) = 1 + \text{tr} \mathbf{A} + O(\|\mathbf{A}\|^2)$, because every term in the expansion other than the product of the diagonal entries is at least quadratic in the entries of $\mathbf{A}$. Hence

$$\det(\mathbf{F} + \delta\mathbf{F}) = J \left[1 + \text{tr}(\mathbf{F}^{-1}\delta\mathbf{F}) \right] + O(\|\delta\mathbf{F}\|^2) = J + J F_{ji}^{-1} \delta F_{ij} + O(\|\delta\mathbf{F}\|^2), \quad (40)$$

and comparing with $\delta J = (\partial J / \partial F_{ij}) \delta F_{ij}$ we obtain

$$\frac{\partial J}{\partial F_{ij}} = J F_{ji}^{-1}, \quad (41)$$

which is Jacobi's formula; note the transposed indices on the right-hand side. The chain rule then gives the first Piola–Kirchhoff stress

$$T_{ij} = \frac{\partial W}{\partial F_{ij}} = \kappa(J-1) \frac{\partial J}{\partial F_{ij}} = \kappa(J-1) J F_{ji}^{-1}, \quad \text{i.e.} \quad \mathbf{T} = \kappa(J-1) J \mathbf{F}^{-T}. \quad (42)$$

For the uniform stretch $\mathbf{F} = \text{diag}(\lambda_1, \lambda_2, \lambda_3)$, $J = \lambda_1 \lambda_2 \lambda_3$ and $\mathbf{F}^{-T}$ is diagonal with entries λ_i^{-1} , so that

$$\mathbf{T} = \kappa(J-1) \text{diag}(\lambda_2 \lambda_3, \lambda_3 \lambda_1, \lambda_1 \lambda_2). \quad (43)$$

(b) On the face $x_1 = 1$ of the reference cube the outward unit normal is $\hat{\mathbf{n}} = (1, 0, 0)$, and from eq. (43) the traction is

$$\mathbf{t} = \mathbf{T} \hat{\mathbf{n}} = \kappa(J-1) \lambda_2 \lambda_3 (1, 0, 0). \quad (44)$$

This is the force per unit *reference* area exerted on the body across the surface. The face has unit area in the reference body, so the total force on its image is $\kappa(J-1) \lambda_2 \lambda_3$ in the x_1 -direction. That image is the face $y_1 = \lambda_1$ of the box M_t , which has current area $\lambda_2 \lambda_3$, so the force per unit current area is

$$\mathbf{t}^{\text{Cauchy}} = \frac{\kappa(J-1) \lambda_2 \lambda_3}{\lambda_2 \lambda_3} (1, 0, 0) = \kappa(J-1) (1, 0, 0), \quad (45)$$

and the two tractions differ by the factor $\lambda_2 \lambda_3 = J / \lambda_1$, the ratio of current to reference area of the face. For $J > 1$ the traction is directed outwards: an outward pull must be applied to hold the body in a dilated state, and conversely a compressed body ($J < 1$) pushes back on whatever confines it.

The same argument applied to the other faces shows that the Cauchy traction on every face is $\kappa(J-1) \hat{\mathbf{n}}_t$, with $\hat{\mathbf{n}}_t$ the current outward normal, so the Cauchy stress is the isotropic tensor $\kappa(J-1) \mathbf{1}$: a body with this strain energy behaves as a compressible fluid with pressure $-\kappa(J-1)$, and for small volume changes $J-1 \approx \nabla \cdot \mathbf{u}$ we recover the linear relation between pressure and dilatation with bulk modulus κ . The general relation between the two stress tensors is $\boldsymbol{\sigma} = J^{-1} \mathbf{T} \mathbf{F}^T$, which indeed reduces to $\kappa(J-1) \mathbf{1}$ for eq. (42); the Piola–Kirchhoff and Cauchy stresses coincide only when $\mathbf{F} = \mathbf{1}$, which is why their distinction can be ignored in linearised theories about a stress-free reference state.